The Use of SBAS GIVE for Improving CAT-I GBAS Availability in the Korean Region

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ABSTRACT

Anomalous ionospheric behaviors cause spatially different refraction effects on the signals of Global Navigation Satellite System (GNSS). This can pose potential integrity threats to Ground-Based Augmentation System (GBAS) users. Very rare ionospheric threats, which are not monitored by range-domain monitors within GBAS ground facility, are mitigated via position-domain geometry screening by inflating some of integrity parameters under the worst-case ionospheric conditions. However, this underlying assumption with a very small probability reduces usable satellite subset geometries in process of inflation, and leads to availability loss of GBAS. This paper utilizes Space-Based Augmentation System (SBAS) Grid Ionospheric Vertical Error (GIVE) to mitigate severe ionospheric threats, and improves the CAT-I GBAS availability in the Korean region. Ionospheric observation data are obtained from 5 South Korean GNSS reference stations for computation of GIVEs, and the GIVEs are used for determination of execution of position-domain geometry screening. In addition, CAT-I GBAS availability is evaluated by computing a 24-hour average of instantaneous availability. The result demonstrates that the use of GIVEs reduces not only unnecessary inflation but also availability degradation. This benefit of using data external to GBAS can reduce the conservatism of worst-case ionospheric assumption for implementation of GBAS concerning availability loss in the future.

Keywords: GBAS, SBAS GIVE, Integrity, Availability, Ionospheric Anomaly Mitigation

1. INTRODUCTION

Ground-Based Augmentation System (GBAS) supports aircraft precision approach and landing by broadcasting differential Global Positioning System (GPS) corrections and integrity information to aviation users. Anomalous ionospheric behaviors cause spatially different refraction effects on the signals of GPS, and ionospheric delay errors experienced by airplane and GBAS ground facility can be largely different. These result in unacceptably large residual errors that are not completely removed by applying differential GPS corrections. In addition, very rare ionospheric threats may not be monitored in range domain by Code-Carrier Divergence (CCD) monitor if the ionospheric front speed is almost close to that of the speed of Ionospheric Pierce Point (IPP) between impacted satellite and GBAS ground facility. *Lee et al.* [2] developed the Position-Domain Geometry Screening (PDGS) method to mitigate worst-case ionospheric threats by inflating broadcast integrity parameters. However, occasionally the underlying assumption reduces the number of usable satellite subset geometries in the process of inflation, and may lead to availability loss of GBAS operation.

This paper mitigates severe ionospheric threats and improves the CAT-I GBAS availability in the Korean region by utilizing Space-Based Augmentation System (SBAS) Grid Ionospheric Vertical Error (GIVE), based on a technique that was originally developed by *Pullen et al.* [3]. Section 2 describes the data used in this paper. In section 3, we review the methodology of ionospheric mitigation scheme using SBAS GIVEs and availability analysis. Section 4 presents the results of CAT-I GBAS availability analysis and section 5 draws a conclusion.

2. DATA

GPS observation data, dual-frequency (L1 at 1575.42MHz and L2 at 1227.60MHz) code and carrier-phase measurements, are obtained from five South Korean GPS reference stations for computation of GIVEs. Figure 1 shows the locations of 5 South Korean GPS reference stations, namely CHJU, CNJU, INCH, KARN, and PUSN. Precise ionospheric delays on the L1 signal are estimated using the Long-Term Ionospheric Anomaly Monitor (LTIAM) [4]. Furthermore, this paper uses the Kriging method defined in [5][6] for estimating GIVE at five-to-five degree Ionospheric Grid Point (IGP) within latitudes of 5-65° and longitudes of 95-165° using the computed precise ionospheric delays.

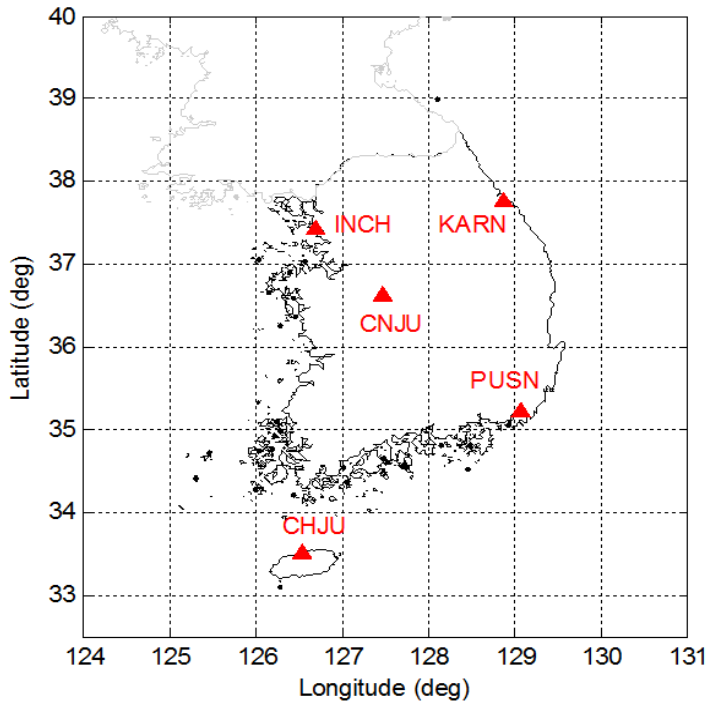

Figure 1. Locations of five South Korean GPS reference stations for SBAS GIVE computation.

3. METHODOLOGY

3.1 GBAS Position-Domain Geometry Screening

As we previously mentioned in section 1, GBAS ground facility mitigates ionospheric anomalies via PDGS, assuming the worst-case ionospheric condition. Very rare ionospheric threats, which are improbable to be detected in range domain by CCD monitor, can produce spatially large decorrelation and pose potential integrity threat to GBAS users. This effect on GBAS users is modeled by calculating Maximum Ionospheric Error in Vertical (MIEV) based on the ionospheric anomaly threat model [2]. These ionospheric-induced errors in position domain are computed over all possible satellite subset geometries for a particular epoch, and are compared to Obstacle Clearance Surface (OCS)-derived criteria, also known as CAT-I Tolerable Error Limit (TEL), in vertical domain (28.78 m at the minimum decision height of 200 ft) to determine potentially hazardous subset geometries [2][8]. Ionospheric anomaly mitigation is made by inflating chosen integrity parameters (σ_{vig} is used in this paper) until Vertical Protection Levels (VPLs) of the potentially hazardous subset geometries exceed the Vertical Alert Limit (VAL) (10 m at the minimum decision height of 200 ft) [2].

Figure 2 shows an example of MIEVs, nominal VPLs, and inflated VPLs for a particular epoch simulated at Gimpo airport to meet integrity requirement under the worst-case ionospheric scenario. The MIEVs, nominal VPLs, and inflated VPLs are denoted by red circles, blue crosses, and blue diamonds. Black solid lines indicate TEL (28.78 m) and VAL (10 m) at CAT-I decision height. As shown in this figure, the MIEV of 7th subset geometry in the red box is larger than TEL, and thus this subset is considered potentially hazardous. Then, σ_{vig} is inflated until the VPL of the subset becomes greater than VAL. Hence, this hazardous geometry is properly removed by applying the inflated σ_{vig} for VPL computation.

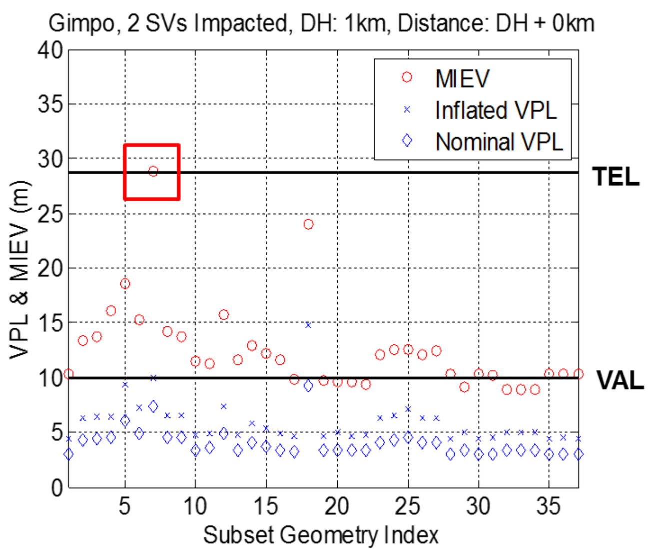

Figure 2. An example of MIEVs, nominal VPLs, inflated VPLs for a particular epoch simulated at Gimpo airport to meet integrity requirement under the worst-case ionospheric scenario. MIEVs, nominal VPLs, and inflated VPLs are denoted by red circles, blue crosses, and blue diamonds. Black solid lines indicate TEL and VAL at CAT-I DH.

3.2 Ionospheric Mitigation using SBAS GIVE Information

SBAS provides users both ionospheric delay estimates and their ionospheric uncertainty bounds, also called GIVE, at each IGP. Because SBAS corrections are computed over several hundreds of kilometers, ionospheric spatial gradient, which can cause severe errors for SBAS users, is not hazardous to GBAS users [3]. In other words, SBAS monitors such as "Extreme Storm Detector," should have detected ionospheric spatial gradients that can pose integrity threats to GBAS users prior to any impacts on GBAS [3][7]. Table 1 shows GBAS classification corresponding to SBAS GIVE values. GIVE values of 0-6 m guarantee no ionospheric threats to GBAS (good case) [3]. GIVE values of 15 m or "Not monitored" indicate that SBAS cannot provide any confirmation and assurance of ionospheric anomalies due to insufficient ionospheric observations (neutral case)

[3]. GIVE value of 45 m informs that a potential ionospheric threat (bad case) exists [3]. GBAS ground facility checks and updates satellite-specific ionospheric status based on the four surrounding GIVE values. If a GBAS subset geometry contains a satellite whose IPP is surrounded by one or more “neutral” or “bad” IGPs, MIEV of this subset geometry is calculated. However, MIEV is not determined for a subset geometry when GBAS IPP is within a triangle formed with three good cases of IGPs even if the fourth IGP is neutral [3].

Table 1. GBAS Classification corresponding SBAS GIVE values [3]

GIVE Value	GBAS Class	Notes
≤ 6.0 m	Good	SBAS verifies that no threat is present.
15.0 m	Not Observed	SBAS observations are too limited to confirm that no threat exists.
45.0 m	Bad	SBAS detects a nearby ionosphere storm – possible threat.
Not monitored	Not Observed	SBAS observations are too limited to provide any ionospheric assurance.

3.3 Availability Analysis

Ionospheric anomaly mitigation simulation via PDGS using SBAS GIVE information for a particular nominal day is carried out at a distance of 6 km from the ground facility which reaches the 200 ft decision height of CAT-I precision approach. The standard RTCA 24 GPS constellation (Table B-1 in [9]) is used, and the GBAS ground facility is assumed to be located at the Gimpo international airport (37.6 °, 128.8 °). To analyze availability, usable satellite subset geometries, which have inflated VPLs below VAL for CAT-I, are counted after applying the inflation factor to the VPLs in process of PDGS at five-minute time intervals. The probability, at which a specific usable subset of satellite geometry may be operating, is calculated using standard constellation state probability, and the operating probability is added to the total instantaneous availability for a particular epoch [9][10]. In addition, CAT-I GBAS availability is evaluated by computing a 24- hour average of the instantaneous availability for a particular nominal sidereal day [10].

4. RESULT

IPP coverage of all GPS satellites tracked by GBAS ground facility at Gimpo airport and 95% of GIVE values in the Korean region for a particular nominal day are shown in Figure 3. IPPs of all GPS satellites tracked at Gimpo airport for 24 hours are within the white dashed rectangular box. The GIVE values in the center of the Korean region are less than or equal to 6 m while those in the edge of the white box are larger than 6.0 m due to lack of SBAS observability. Many of GBAS satellite subset geometries contain the GBAS IPPs in this edge of the white box (neutral case) and these result in MIEV calculations and comparisons with TEL.

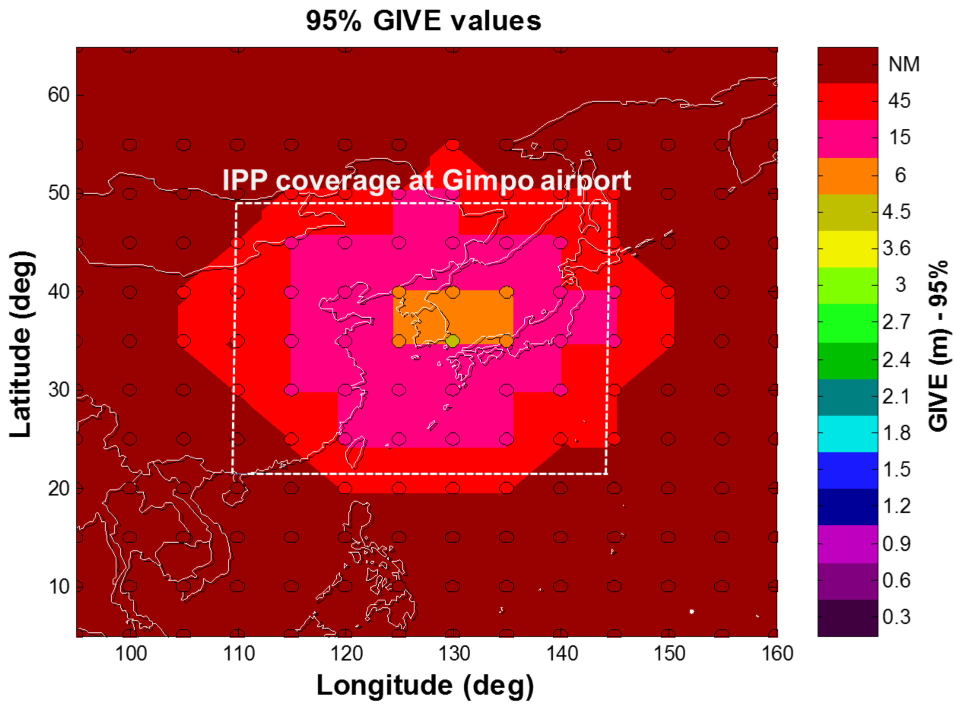

Figure 3. IPP coverage of all GPS satellites tracked by GBAS ground facility at Gimpo airport and 95% of GIVE values in the Korean region. IPPs of all GPS satellites tracked at Gimpo airport for 24 hours are within the white dashed rectangular box.

Figure 4 shows the numbers of unusable satellite subset geometries for 24 hours when the PDGS stand-alone (blue-dashed star) and the PDGS using SBAS GIVE (red-dashed circle) were applied, respectively. Unusable subset geometry is counted if its inflated VPL exceeds VAL at five-minute time intervals. The total number of unusable subset geometries decreases from 1350 to 1270 as summarized in Table 2. This result implies that from the SBAS GIVE information, many of subset geometries are confirmed to have no ionospheric threats. CAT-I GBAS availability for each case is evaluated as 99.913348 % (PDGS stand-alone) and 99.919701 % (PDGS using SBAS GIVE). This suggests that the use of external information for GBAS results in availability improvement.

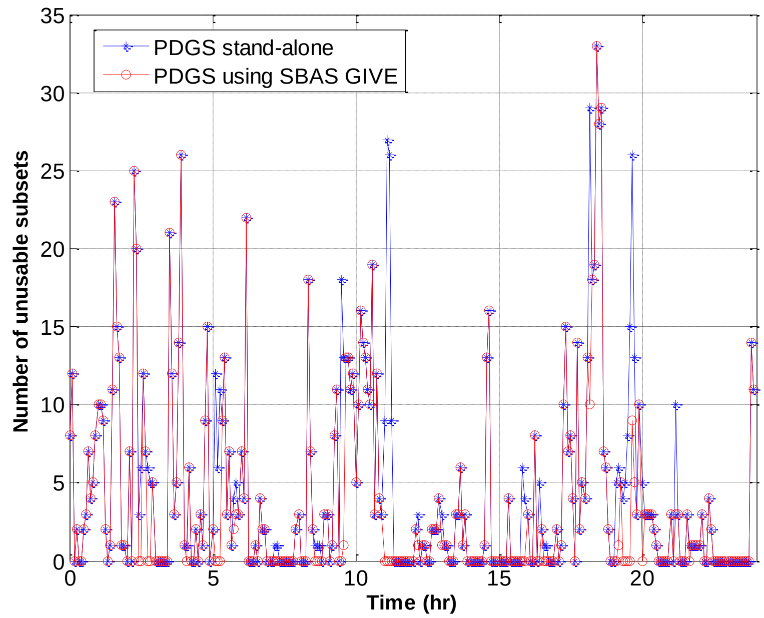

Figure 4. The numbers of unusable satellite subset geometries for the PDGS stand-alone (blue-dashed star) and the PDGS using SBAS GIVE (red-dashed circle). Unusable subset geometry is counted if its inflated VPL exceeds VAL at five-minute time intervals.

Table 2. The number of unusable satellite subset geometries and availability for the PDGS stand-alone and the PDGS using SBAS GIVE. Unusable subset geometry is counted if its inflated VPL exceeds VAL at five-minute time intervals.

	PDGS stand-alone	PDGS using SBAS GIVE
Number of unusable satellite subset geometries	1350	1070
Availability (%)	99.913348	99.919701

5. CONCLUSION

This paper demonstrates availability improvement of the CAT-I GBAS in the Korean region when SBAS GIVE information is used. Many subset geometries were not discarded, benefitted from using SBAS GIVE information, during PDGS process, and this led to the reduction of unusable satellite subset geometries by approximately 20.7% for 24 hours (288 evaluating points). CAT-I GBAS availability increased for a particular nominal day, and this indicates that a large portion of satellite subset geometries can be unconstrained by potential ionospheric anomalies with SBAS GIVE information.

We concentrated on taking the benefit of using SBAS GIVEs with the current CAT-I GBAS, based on the simple inflation technique, and this reduced the conservatism of worst-case ionospheric scenario outlined in this paper. However, further improvement on the strategy of integrity parameter inflation and MIEV calculation in PDGS with SBAS GIVE will be needed in order to minimize the availability loss from the worst-case ionospheric assumption.

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